



National
Qualifications
2015

2015 Mathematics

New Higher Paper 1

Finalised Marking Instructions

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General Comments

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For each question the marking instructions are in two sections, namely **Illustrative Scheme** and **Generic Scheme**. The **Illustrative Scheme** covers methods which are commonly seen throughout the marking. The **Generic Scheme** indicates the rationale for which each mark is awarded. In general, markers should use the **Illustrative Scheme** and only use the **Generic Scheme** where a candidate has used a method not covered in the **Illustrative Scheme**.

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- 3 Working subsequent to an error **must be followed through**, with possible full marks for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working for a follow through mark has been eased, the follow through mark cannot be awarded.
- 4 As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Throughout this paper, unless specifically mentioned in the marking instructions, a correct answer with no working receives no credit.
- 5 In general, as a consequence of an error perceived to be trivial, casual or insignificant, e.g. $6 \times 6 = 12$, candidates lose the opportunity of gaining a mark. But note the second example in comment 7.
- 6 Where a transcription error (paper to script or within script) occurs, the candidate should be penalised, eg

The diagram illustrates three scenarios for marking a transcription error in a quadratic equation. Each scenario is shown in a box with a callout explaining the marking decision.

Scenario 1: The box shows the equations $x^2 + 5x + 7 = 9x + 4$, $x - 4x + 3 = 0$, and $x = 1$. The first equation is marked with a red checkmark. The second equation is marked with a red X. The third equation is marked with a red checkmark and a '2' in a box. A callout points to the second equation, stating: "This is a transcription error and so the mark is not awarded."

Scenario 2: The box shows the equations $x^2 + 5x + 7 = 9x + 4$, $x - 4x + 3 = 0$, and $x = 1$. The first equation is marked with a red checkmark. The second equation is marked with a red X. The third equation is marked with a red checkmark and a '2' in a box. A callout points to the second equation, stating: "Eased as no longer a solution of a quadratic equation."

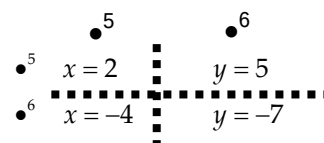
Scenario 3: The box shows the equations $x^2 + 5x + 7 = 9x + 4$, $x - 4x + 3 = 0$, and $(x - 3)(x - 1) = 0$. The first equation is marked with a red checkmark. The second equation is marked with a red checkmark. The third equation is marked with a red checkmark. A callout points to the second equation, stating: "Exceptionally this error is not treated as a transcription error as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt."

7 Vertical/horizontal marking

Where a question results in two pairs of solutions, this technique should be applied, but only if indicated in the detailed marking instructions for the question.

Example: Point of intersection of line with curve

Illustrative Scheme: •⁵ $x = 2, x = -4$
•⁶ $y = 5, y = -7$



Markers should choose whichever method benefits the candidate, but **not** a combination of both.

8 In final answers, numerical values should be simplified as far as possible, unless specifically mentioned in the detailed marking instructions.

Examples: $\frac{15}{12}$ should be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ should be simplified to 43

$\frac{15}{0.3}$ should be simplified to 50

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$\sqrt{64}$ must be simplified to 8

The square root of perfect squares up to and including 100 must be known.

9 Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

10 Unless specifically mentioned in the marking instructions, the following should not be penalised:

- Working subsequent to a **correct** answer;
- Correct working in the wrong part of a question;
- Legitimate variations in numerical answers, eg angles in degrees rounded to nearest degree;
- Omission of units;
- Bad form (bad form only becomes bad form if subsequent working is correct), e.g.
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$
written as
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$
 $2x^4 + 4x^3 + 6x^2 + 4x + x^3 + 2x^2 + 3x + 2$
 $2x^4 + 5x^3 + 8x^2 + 7x + 2$ gains full credit;
- Repeated error within a question, but not between questions.

11 In any 'Show that . . .' question, where the candidate has to arrive at a required result, the last mark of that part is not available as a follow through from a previous error unless specifically stated otherwise in the detailed marking instructions.

- 12** All working should be carefully checked, even where a fundamental misunderstanding is apparent early in the candidate's response. Marks may still be available later in the question so reference must be made continually to the marking instructions.
All working must be checked: the appearance of the correct answer does not necessarily indicate that the candidate has gained all the available marks.
- 13** If you are in serious doubt whether a mark should or should not be awarded, consult your Team Leader (TL).
- 14** Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.
- 15** Where a candidate has made multiple attempts using the same strategy, mark all attempts and award the lowest mark.
Where a candidate has tried different strategies, apply the above ruling to attempts within each strategy and then award the highest resultant mark. For example:

Strategy 1 attempt 1 is worth 3 marks	Strategy 2 attempt 1 is worth 1 mark
Strategy 1 attempt 2 is worth 4 marks	Strategy 2 attempt 2 is worth 5 marks
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

- 16** In cases of difficulty, covered neither in detail nor in principle in these instructions, markers should contact their TL in the first instance.

Detailed Marking Instructions for each question

Question		Generic Scheme	Illustrative Scheme	Max Mark
1.				
		•¹ equate scalar product to zero •² state value of t	•¹ $-24 + 2t + 6 = 0$ •² $t = 9$	2
Notes:				
Commonly Observed Responses:				
Candidate A $-24 + 2t + 6 = -1$ • ¹ × $t = \frac{17}{2}$ or $8\frac{1}{2}$ • ² ✓1				
2.				
		•¹ know to and differentiate •² evaluate $\frac{dy}{dx}$ •³ evaluate y-coordinate •⁴ state equation of tangent	•¹ $6x^2$ •² 24 •³ -13 •⁴ $y = 24x + 35$	4
Notes:				
1. • ⁴ is only available if an attempt has been made to find the gradient from differentiation. 2. At mark • ⁴ accept $y + 13 = 24(x + 2)$, $y - 24x = 35$ or any other rearrangement of the equation.				
Commonly Observed Responses:				

Question	Generic Scheme	Illustrative Scheme	Max Mark
3.			
	•¹ know to use $x = -3$ •² interpret result and state conclusion •³ state quadratic factor •⁴ factorise completely	Method 1 •¹ $(-3)^3 - 3(-3)^2 - 10(-3) + 24$ •² $= 0 \therefore (x + 3)$ is a factor. Method 2 •¹ $\begin{array}{r rrrr} -3 & 1 & -3 & -10 & 24 \\ & & -3 & & \\ \hline & 1 & & & \end{array}$ •² $\begin{array}{r rrrr} -3 & 1 & -3 & -10 & 24 \\ & & -3 & 18 & -24 \\ \hline & 1 & -6 & 8 & 0 \end{array}$ remainder = 0 $\therefore (x + 3)$ is a factor. Method 3 •¹ $\begin{array}{r} x^2 \\ x+3 \overline{) x^3 - 3x^2 - 10x + 24} \\ \underline{x^3 + 3x^2} \end{array}$ •² $= 0 \therefore (x + 3)$ is a factor. •³ $x^2 - 6x + 8$ stated or implied by •⁴ •⁴ $(x + 3)(x - 4)(x - 2)$	4

Notes:

- Communication at •² must be consistent with working at that stage ie a candidate's working must arrive legitimately at 0 before •² is awarded.
- Accept any of the following for •²:
 ' $f(-3) = 0$ so $(x + 3)$ is a factor'
 'since remainder is 0, it is a factor'
 the 0 from the table linked to the word 'factor' by eg 'so', 'hence', ' $\therefore$ ', ' $\rightarrow$ ', ' $\Rightarrow$ '
- Do not accept any of the following for •²:
 double underlining the zero or boxing the zero without comment
 ' $x = 3$ is a factor', ' $(x - 3)$ is a factor', ' $x = -3$ is a root', ' $(x - 3)$ is a root', " $(x + 3)$ is a root"
 the word 'factor' **only**, with no link
- At •⁴ the expression may be written in any order.
- An incorrect quadratic correctly factorised may gain •⁴
- Where the quadratic factor obtained is irreducible, candidates must clearly demonstrate that $b^2 - 4ac < 0$ to gain •⁴
- $= 0$ must appear at •¹ or •² for •² to be awarded.
- For candidates who do not arrive at 0 at the •² stage •²•³•⁴ not available.
- Do not penalise candidates who attempt to solve a cubic equation. However, within this working there may be evidence of the correct factorisation of the cubic.

Commonly Observed Responses:			
Candidate A		Candidate B	
$\begin{array}{r} 2 \ 1 \ -3 \ -10 \ 24 \\ \underline{ } \\ 2 \ -2 \ -24 \end{array}$ $1 \ -1 \ -12 \ 0 \Rightarrow x-2 \text{ is a factor}$ $(x-2)(x^2-x-12)$ $(x-2)(x-4)(x+3) \Rightarrow x+3 \text{ is a factor}$ •² ✓1 •³ ✓ •⁴ ✓ •¹ ✓		$\begin{array}{r} 2 \ 1 \ -3 \ -10 \ 24 \\ \underline{ } \\ 2 \ -2 \ -24 \end{array}$ $1 \ -1 \ -12 \ 0 \Rightarrow x-2 \text{ is a factor}$ •¹ ✓2 •² ✓1	
4.			
	•¹ state the value of p •² state the value of q •³ state the value of r	•¹ $p = 3$ •² $q = 4$ •³ $r = 1$	3
Notes:			
1. These are the only acceptable responses for p , q and r .			
Commonly Observed Responses:			
5(a).			
	•¹ let $y = 6 - 2x$ and rearrange. •² state expression. Method 2 •³ equates composite function to x •¹ start to rearrange. •² state expression.	•¹ $x = \frac{6-y}{2}$ or $y = \frac{6-x}{2}$ •² $g^{-1}(x) = \frac{6-x}{2}$ or $3 - \frac{x}{2}$ or $\frac{x-6}{-2}$ Method 2 $g(g^{-1}(x)) = x$ this gains •³ $6 - 2g^{-1}(x) = x$ $g^{-1}(x) = \frac{6-x}{2}$ or $3 - \frac{x}{2}$ or $\frac{x-6}{-2}$	2
Notes:			
1. At • ¹ accept any equivalent expression with any 2 distinct variables.			
Commonly Observed Responses:			
5(b).			
	• ³ state expression	• ³ x	1
Notes:			
2. Candidates using method 2 may be awarded • ³ at line one.			
3. For candidates who attempt to find the composite function $g(g^{-1}(x))$, accept $6 - 2\left(\frac{6-x}{2}\right)$ for • ³ .			
4. In this case • ³ may be awarded as follow through where an incorrect $g^{-1}(x)$ is found at • ² , provided it includes the variable x .			
Commonly Observed Responses:			

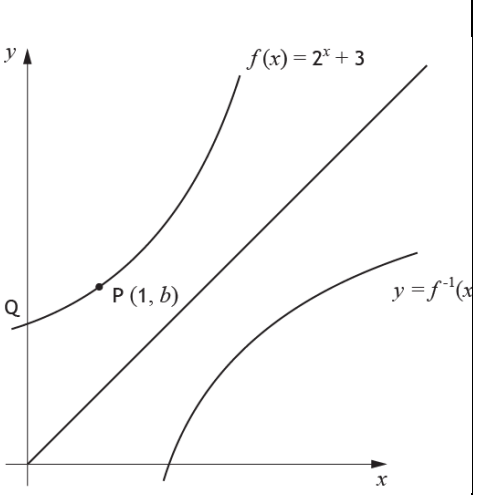
Question		Generic Scheme	Illustrative Scheme	Max Mark
6.				
		•¹ use laws of logs •² use laws of logs •³ evaluate log	•¹ $\log_6 27^{\frac{1}{3}}$ •² $\log_6 \left(12 \times 27^{\frac{1}{3}} \right)$ •³ 2	3
Notes:				
Commonly Observed Responses:				
Candidate A		Candidate B		
$\log_6 12 + \log_6 9$		$\frac{1}{3} \log_6 (12 \times 27)$		
$\log_6 (12 \times 9)$		$\frac{1}{3} \log_6 324$		
$\log_6 108$		$\log_6 324^{\frac{1}{3}}$		
		Award 1 out of 3 ^,^ ☒		
7.				
		•¹ write in differentiable form •² differentiate first term •³ differentiate second term •⁴ evaluate derivative at $x = 4$	•¹ $3x^{\frac{3}{2}} - 2x^{-1}$ •² $\frac{9}{2}x^{\frac{1}{2}} + \dots$ •³ $\dots + 2x^{-2}$ •⁴ $9\frac{1}{8}$	4
Notes:				
1. •² must involve a fractional index. 2. •³ must involve a negative index. 3. •⁴ is only available as a consequence of substituting into a 'derivative' containing a fractional or negative index. 4. If no attempt has been made to expand the bracket at •¹ then •²&•³ are not available. •⁴ is still available as follow through.				
Commonly Observed Responses:				
Candidate A				
$f(x) = 3x^{\frac{1}{2}} - 2x^{-\frac{1}{4}}$				
$f'(x) = \frac{3}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{5}{4}}$				
$= \frac{3}{2\sqrt{x}} + \frac{1}{2\sqrt[4]{x^5}}$				
$f'(4) = \frac{3}{2\sqrt{4}} + \frac{1}{2\sqrt[4]{4^5}}$				
$= \frac{3}{4} + \frac{1}{8\sqrt{2}}$				

Question		Generic Scheme	Illustrative Scheme	Max Mark
8.				
		¹ interpret information ² express in standard quadratic form ³ factorise ⁴ state range	¹ $x(x-2) < 15$ ² $x^2 - 2x - 15 < 0$ ³ $(x-5)(x+3) < 0$ ⁴ $2 < x < 5$	4
Notes:				
Commonly Observed Responses:				
Candidate A $x(x-2) = 15$ $x^2 - 2x - 15 = 0$ $x = -3, 5$		¹ ✗ ² ✓ 2 ³ ✓ 1 ⁴ ^	Candidate B - Mistaking perimeter for area $4x - 4 < 15$ $x < \frac{19}{4}$ Award 1/4	
Candidate C $x^2 - 2x < 15$ $x > 2$ Award 1/4			Candidate D $x^2 - 2x < 15$ $x > 2$ $x < 5$ Award 2/4	Inequalities not linked by 'and'
Candidate E $x^2 - 2x < 15$ $x > 2$ and $x < 5$ Award 4/4		Inequalities linked by 'and'		

Question		Generic Scheme	Illustrative Scheme	Max Mark
9.				
		•¹ find gradient of AB •² calculate gradient of BC •³ interpret results and state conclusion	•¹ $m_{AB} = -\sqrt{3}$ •² $m_{BC} = -\frac{1}{\sqrt{3}}$ •³ $m_{AB} \neq m_{BC} \Rightarrow$ points are not collinear. Method 2 •¹ $m_{AB} = -\sqrt{3}$ •² AB makes 120° with positive direction of the x-axis. •³ $120 \neq 150$ so points are not collinear.	3
Notes:				
1. The statement made at • ³ must be consistent with the gradients or angles found for • ¹ and • ² .				
Commonly Observed Responses:				
10(a).				
		• ¹ state value of $\cos 2x$	• ¹ $\frac{4}{5}$	1
Notes:				
Commonly Observed Responses:				
Candidate A		Candidate B		
$\cos 2x = \frac{3}{5}$ •¹ $\times$ •² ✓1 •³ ✓1 $2\cos^2 x - 1 = \dots$ $\cos x = \frac{2}{\sqrt{5}}$		$\cos 2x = 4$ •¹ $\times$ •² ✓1 $2\cos^2 x - 1 = 4$ $\cos^2 x = \frac{5}{2}$ $\cos x = \sqrt{\frac{5}{2}}$ •³ $\times$ invalid answer		
10(b).				
		•² use double angle formula •³ evaluate $\cos x$	•² $2\cos^2 x - 1 = \dots$ •³ $\frac{3}{\sqrt{10}}$	2
Notes:				
1. Ignore the inclusion of $-\frac{3}{\sqrt{10}}$. 2. At • ² the double angle formula must be equated to the candidates answer to part (a).				
Commonly Observed Responses:				

Question		Generic Scheme	Illustrative Scheme	Max Mark
11(a).				
		•¹ state coordinates of centre •² find gradient of radius •³ state perpendicular gradient •⁴ determine equation of tangent	•¹ $(-8, -2)$ •² $-\frac{1}{2}$ •³ 2 •⁴ $y = 2x - 1$	4
Notes:				
1. •⁴ is only available as a consequence of trying to find and use a perpendicular gradient. 2. At mark •⁴ accept $y + 5 = 2(x + 2)$, $y - 2x = -1$, $y - 2x + 1 = 0$ or any other rearrangement of the equation.				
Commonly Observed Responses:				

Question	Generic Scheme	Illustrative Scheme	Max Mark
11(b).			
	Method 1 •⁵ arrange equation of tangent in appropriate form and equate y_{tangent} to y_{parabola} •⁶ rearrange and equate to 0 •⁷ know to use discriminant and identify a, b, and c •⁸ simplify and equate to 0 •⁹ start to solve •¹⁰ state value of p Method 2 •⁵ arrange equation of tangent in appropriate form and equate y_{tangent} to y_{parabola} •⁶ find $\frac{dy}{dx}$ for parabola •⁷ equate to gradient of line and rearrange for p •⁸ substitute and arrange in standard form •⁹ factorise and solve for x •¹⁰ state value of p	Method 1 •⁵ $2x - 1 = -2x^2 + px + 1 - p$ •⁶ $2x^2 + (2 - p)x + p - 2 = 0$ •⁷ $(2 - p)^2 - 4 \times 2 \times (p - 2)$ •⁸ $p^2 - 12p + 20 = 0$ •⁹ $(p - 10)(p - 2) = 0$ •¹⁰ $p = 10$ Method 2 •⁵ $2x - 1 = -2x^2 + px + 1 - p$ •⁶ $\frac{dy}{dx} = -4x + p$ •⁷ $2 = -4x + p$ $p = 2 + 4x$ •⁸ $0 = 2x^2 - 4x$ •⁹ $0 = 2x(x - 2)$ $x = 0, x = 4$ •¹⁰ $p = 10$	6
Notes:			
1. At • ⁶ accept $2x^2 + 2x - px + p - 2 = 0$.			
2. At • ⁷ accept $a = 2$, $b = (2 - p)$, and $c = (p - 2)$.			
Commonly Observed Responses:			
Just using the parabola $a = -2 \quad b = p \quad c = 1 - p$ $b^2 - 4ac = p^2 - 4 \times (-2)(1 - p)$ $= p^2 - 8p + 8 = 0$ $p = 4 \pm \sqrt{8}$ $p = 4 + \sqrt{8} \text{ as } p > 3$ •⁵ $\wedge$ •⁶ $\wedge$ •⁷ ☒ 1 •⁸ ☒ 2 •⁹ ☒ 1 •¹⁰ ☒ 1			

Question		Generic Scheme	Illustrative Scheme	Max Mark
12.				
		¹ interpret integral below x-axis ² evaluate	¹ -1 (accept area below x-axis = 1) ² $-\frac{1}{2}$	2
Notes:				
1. For candidates who calculate the area as $\frac{3}{2}$ award 1 out of 2.				
Commonly Observed Responses:				
13(a)				
		¹ calculate b	¹ 5	1
Notes:				
Commonly Observed Responses:				
13 (b)(i)				
		² reflecting in the line $y = x$	² 	1
Notes:				
1. If the reflected graph cuts the y -axis, ² is not awarded.				
Commonly Observed Responses:				

Question		Generic Scheme	Illustrative Scheme	Max Mark						
13(b)(ii)										
		•³ calculate y intercept •⁴ state coordinates of image of Q •⁵ state coordinates of image of P	•³ 4 •⁴ (4, 0) see note 2 •⁵ (5, 1)	3						
Notes:										
2. •⁴ can only be awarded if (4,0) is clearly identified either by their labelling or by their diagram. 3. •³ is awarded for the appearance of 4, or (4,0) or (0,4) . 4. •⁵ is awarded for the appearance of (5,1) . Ignore any labelling attached to this point.										
Commonly Observed Responses:										
Candidate A		Candidate B								
$y = f(x)$ reflected in x – axis		$y = f(x)$ reflected in y – axis								
4	• ³ ✓	4	• ³ ✓							
(0,-4)	• ⁴ ✓ 2	(0,4)	• ⁴ ✓ 2							
(1,-5)	• ⁵ ✓ 1	(-1,5)	• ⁵ ✓ 2							
13(c)										
		•⁶ state x coordinate of R •⁷ state y coordinate of R	•⁶ $x = 2$ •⁷ $y = -7$	2						
Notes:										
Commonly Observed Responses:										
14.										
		•¹ identify length of radius •² determine value of k	<table><tr><td>y – axis tangent to circle</td><td>Circle passes through origin</td></tr><tr><td>•¹ $r = 6$</td><td>$r = \sqrt{61}$</td></tr><tr><td>•² $k = 25$</td><td>$k = 0$</td></tr></table>	y – axis tangent to circle	Circle passes through origin	• ¹ $r = 6$	$r = \sqrt{61}$	• ² $k = 25$	$k = 0$	2
y – axis tangent to circle	Circle passes through origin									
• ¹ $r = 6$	$r = \sqrt{61}$									
• ² $k = 25$	$k = 0$									

Question		Generic Scheme	Illustrative Scheme	Max Mark
15.				
		•¹ know to integrate •² integrate a term •³ complete integration •⁴ find constant of integration •⁵ find value of k •⁶ state expression for T	•¹ $\int$ •² $\frac{1}{50}t^2 \dots$ or $\dots - kt$ •³ $\dots - kt$ or $\frac{1}{50}t^2 \dots$ •⁴ $c = 100$ •⁵ $k = 2$ •⁶ $T = \frac{1}{50}t^2 - 2t + 100$	6
Notes:				
1. Accept unsimplified expressions at • ² and • ³ stage. 2. • ⁴ , • ⁵ and • ⁶ are not available for candidates who have not considered the constant of integration. 3. • ¹ may be implied by • ² .				
Commonly Observed Responses:				

[END OF MARKING INSTRUCTIONS]



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- 6 Where a transcription error (paper to script or within script) occurs, the candidate should be penalised, e.g.

This is a transcription error and so the mark is not awarded.

Eased as no longer a solution of a quadratic equation.

Exceptionally this error is not treated as a transcription error as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt.

$$x^2 + 5x + 7 = 9x + 4 \quad \checkmark$$

$$x - 4x + 3 = 0 \quad \times$$

$$x = 1 \quad \boxed{\checkmark 2}$$

$$x^2 + 5x + 7 = 9x + 4 \quad \checkmark$$

$$\text{~}x - 4x + 3 = 0 \quad \checkmark$$

$$(x-3)(x-1) = 0$$

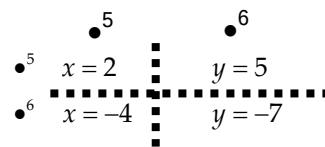
$$x = 1 \text{ or } 3 \quad \checkmark$$

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Example: Point of intersection of line with curve

Illustrative Scheme: $\bullet^5 \quad x = 2, x = -4$
 $\bullet^6 \quad y = 5, y = -7$



Markers should choose whichever method benefits the candidate, but **not** a combination of both.

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Examples: $\frac{15}{12}$ should be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ should be simplified to 43

$\frac{15}{0.3}$ should be simplified to 50

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 - Legitimate variations in numerical answers, eg angles in degrees rounded to nearest degree;
 - Omission of units;
 - Bad form (bad form only becomes bad form if subsequent working is correct), e.g.
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$
written as
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$
 $2x^4 + 4x^3 + 6x^2 + 4x + x^3 + 2x^2 + 3x + 2$
 $2x^4 + 5x^3 + 8x^2 + 7x + 2$ gains full credit;
 - Repeated error within a question, but not between questions.
- 11 In any 'Show that . . .' question, where the candidate has to arrive at a required result, the last mark of that part is not available as a follow through from a previous error unless specifically stated otherwise in the detailed marking instructions.

- 12** All working should be carefully checked, even where a fundamental misunderstanding is apparent early in the candidate's response. Marks may still be available later in the question so reference must be made continually to the marking instructions.
All working must be checked: the appearance of the correct answer does not necessarily indicate that the candidate has gained all the available marks.
- 13** If you are in serious doubt whether a mark should or should not be awarded, consult your Team Leader (TL).
- 14** Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.
- 15** Where a candidate has made multiple attempts using the same strategy, mark all attempts and award the lowest mark.
Where a candidate has tried different strategies, apply the above ruling to attempts within each strategy and then award the highest resultant mark. For example:

Strategy 1 attempt 1 is worth 3 marks	Strategy 2 attempt 1 is worth 1 mark
Strategy 1 attempt 2 is worth 4 marks	Strategy 2 attempt 2 is worth 5 marks
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

- 16** In cases of difficulty, covered neither in detail nor in principle in these instructions, markers should contact their TL in the first instance.

Paper 2

Question	Generic Scheme	Illustrative Scheme	Max Mark
1(a)			
•¹ calculate gradient of AB •² use property of perpendicular lines •³ substitute into general equation of a line •⁴ demonstrate result	•¹ $m_{AB} = -3$ •² $m_{alt} = \frac{1}{3}$ •³ $y - 3 = \frac{1}{3}(x - 13)$ •⁴ $\dots \Rightarrow x - 3y = 4$	4	
Notes:			
1. • ³ is only available as a consequence of trying to find and use a perpendicular gradient. 2. • ⁴ is only available if there is/are appropriate intermediate lines of working between • ³ and • ⁴ . 3. The ONLY acceptable variations for the final equation for the line in • ⁴ are $4 = x - 3y$, $-3y + x = 4$, $4 = -3y + x$.			
Commonly Observed Responses:			
Candidate A $m_{AB} = \frac{-1 - (-5)}{-5 - 7} = \frac{4}{-12} = -\frac{1}{3}$ $m_{alt} = 3$ $y - 3 = 3(x - 13)$ • ⁴ is not available		Candidate B For • ⁴ $y - 3 = \frac{1}{3}x - \frac{13}{3}$ $y = \frac{1}{3}x - \frac{4}{3}$ $3y = x - 4$ - not acceptable $3y - x = -4$ - not acceptable $x - 3y = 4$ ✓	

Question	Generic Scheme	Illustrative Scheme	Max Mark
1(b)			
•⁵ calculate midpoint of AC •⁶ calculate gradient of median •⁷ determine equation of median	•⁵ $M_{AC} = (4, 5)$ •⁶ $m_{BM} = 2$ •⁷ $y = 2x - 3$	3	
Notes:			
4. •⁶ and •⁷ are not available to candidates who do not use a midpoint. 5. •⁷ is only available as a consequence of using a non-perpendicular gradient and a midpoint. 6. Candidates who find either the median through A or the median through C or a side of the triangle gain 1 mark out of 3. 7. At •⁷ accept $y - (-5) = 2(x - (-1))$, $y - 5 = 2(x - 4)$, $y - 2x + 3 = 0$ or any other rearrangement of the equation.			
Commonly Observed Responses:			
Median through A $M_{BC} = (6, -1)$ $m_{AM} = \frac{-8}{11}$ $y + 1 = \frac{-8}{11}(x - 6)$ or $y - 7 = \frac{-8}{11}(x + 5)$ Award 1/3	Median through C $M_{AB} = (-3, 1)$ $m_{CM} = \frac{1}{8}$ $y - 3 = \frac{1}{8}(x - 13)$ or $y - 1 = \frac{1}{8}(x + 3)$ Award 1/3		
1(c)			
•⁸ calculate x or y coordinate •⁹ calculate remaining coordinate of the point of intersection	•⁸ $x = 1$ or $y = -1$ •⁹ $y = -1$ or $x = 1$	2	
Notes:			
8. If the candidate's 'median' is either a vertical or horizontal line then award 1 out of 2 if both coordinates are correct, otherwise award 0.			
Commonly Observed Responses:			
For candidates who find the altitude through B in part (b) $x = -\frac{1}{5}$ $y = -\frac{7}{5}$ • ⁸ ☒ 1 • ⁹ ☒ 1	Candidate A (b) $y - 5 = 2(x - 4)$ • ⁷ ✓ $y = 2x - 13$ -error (c) $x - 3y = 4$ $y = 2x - 13$ Leading to $x = 7$ and $y = 1$ • ⁸ × • ⁹ ☒ 1		

Question	Generic Scheme	Illustrative Scheme	Max Mark
2 (a)			
•¹ interpret notation •² state a correct expression	•¹ $f((1+x)(3-x)+2)$ stated or implied by •² •² $10+(1+x)(3-x)+2$ stated or implied by •³	2	
Notes:			
1. • ¹ is not available for $g(f(x))=g(10+x)$ but • ² may be awarded for $(1+10+x)(3-(10+x))+2$.			
Commonly Observed Responses:			
Candidate A		Candidate B	
(a) $f(g(x)) = 'g(f(x))'$ $= (1+10+x)(3-(10+x))+2$ (b) $= -75-18x-x^2$ or $-x^2-18x-75$ $= -(x^2+18x$ $= -(x+9)^2$ $= -(x+9)^2+6$ (c) $-(x+9)^2+6=0$ $x = -9+\sqrt{6}$ or $-9-\sqrt{6}$		$f(g(x))$ $= 10((1+x)-(3-x))+2$ Candidate C $f(g(x))$ $= 10((1+x)(3-x)+2)$	
•¹ ✗ •² ✓1 •³ ✓1 •⁴ ✓1 •⁵ ✓1 •⁶ ✓1 •⁷ ✓1		•¹ ^ •² ✗ •¹ ^ •² ✗	
2 (b)			
•³ write $f(g(x))$ in quadratic form •⁴ identify common factor •⁵ complete the square	•³ $15+2x-x^2$ or $-x^2+2x+15$ •⁴ $-1(x^2-2x$ stated or implied by •⁵ •⁵ $-1(x-1)^2+16$		
Method 1		Method 1	
Method 2		Method 2	
•⁴ expand completed square form and equate coefficients •⁵ process for q and r and write in required form	•⁴ $px^2+2pqx+pq^2+r$ and $p=-1$, •⁵ $q=-1$ and $r=16$ Note if $p=1$ •⁵ is not available		
3			

Notes:		
2. Accept $16 - (x-1)^2$ or $-[(x-1)^2 - 16]$ at $\bullet^5$.		
Commonly Observed Responses:		
Candidate A $-(x^2 - 2x - 15)$ $\bullet^4$ ✓ $-(x^2 - 2x + 1 - 1 - 15)$ $-(x-1)^2 - 16$ $\bullet^5$ ✗	Candidate B $15 + 2x - x^2$ $\bullet^3$ ✓ $x^2 - 2x - 15$ $\bullet^4$ ✗ $px^2 + 2pqx + pq^2 + r$ and $p=1$ $q=-1$ $r=-16$ $\bullet^5$ ✓ 2 eased	Candidate C $-x^2 + 2x + 15$ $\bullet^3$ ✓ $-(x+1)^2 \dots$ $\bullet^4$ ✗ $-(x+1)^2 + 14$ $\bullet^5$ ✗
Candidate D $15 + 2x - x^2$ $\bullet^3$ ✓ $x^2 - 2x - 15$ $\bullet^4$ ✗ $(x-1)^2 - 16$ $\bullet^5$ ✓ 2 eased Eased, unitary coefficient of x^2 (lower level skill)	Candidate E $15 + 2x - x^2$ $\bullet^3$ ✓ $x^2 - 2x - 15$ $\bullet^4$ ✓ $(x-1)^2 - 16$ so $15 + 2x - x^2 = -(x-1)^2 + 16$ $\bullet^5$ ✓	Candidate F $-x^2 + 2x + 15$ $\bullet^3$ ✓ $-(x+1)^2 \dots$ $\bullet^4$ ✗ $-(x+1)^2 + 16$ $\bullet^5$ ✓ 1
2(c)		
$\bullet^6$ identify critical condition $\bullet^7$ identify critical values	$\bullet^6 -1(x-1)^2 + 16 = 0$ or $f((g(x))) = 0$ $\bullet^7$ 5 and -3	2
Notes:		
3. Any communication indicating that the denominator cannot be zero gains $\bullet^6$.		
4. Accept $x=5$ and $x=-3$ or $x \neq 5$ and $x \neq -3$ at $\bullet^7$.		
5. If $x=5$ and $x=-3$ appear without working award 1/2.		
Commonly Observed Responses:		
Candidate A $\frac{1}{-(x-1)^2 + 16}$ $\bullet^6$ ✓ $x \neq 5$ $\bullet^7$ ✗	Candidate B $\frac{1}{f(g(x))}$ $f(g(x)) > 0$ $\bullet^6$ ✗ $x = -3, x = 5$ $\bullet^7$ ✓ $-3 < x < 5$	
3(a)		
$\bullet^1$ determine the value of the required term	$\bullet^1$ $22\frac{3}{4}$ or $\frac{91}{4}$ or 22.75	1
Notes:		
1. Do not penalise the inclusion of incorrect units.		
2. Accept rounded and unsimplified answers following evidence of correct substitution.		
Commonly Observed Responses:		

Question	Generic Scheme	Illustrative Scheme	Max Mark
3 (b)			
	Method 1 (Considering both limits) •² know how to calculate limit •³ know how to calculate limit •⁴ calculate limit •⁵ calculate limit •⁶ interpret limits and state conclusion Method 2 (Frog first then numerical for toad) •² know how to calculate limit •³ calculate limit •⁴ determine the value of the highest term less than 50 •⁵ determine the value of the lowest term greater than 50 •⁶ interpret information and state conclusion Method 3 (Numerical method for toad only) •² continues numerical strategy •³ exact value •⁴ determine the value of the highest term less than 50 •⁵ determine the value of the lowest term greater than 50 •⁶ interpret information and state conclusion Method 4 (Limit method for toad only) •² & •³ know how to calculate limit •⁴ & •⁵ calculate limit •⁶ interpret limit and state conclusion	Method 1 •² $\frac{32}{1-\frac{1}{3}}$ or $L = \frac{1}{3}L + 32$ •³ $\frac{13}{1-\frac{3}{4}}$ or $L = \frac{3}{4}L + 13$ •⁴ 48 •⁵ 52 •⁶ $52 > 50 \therefore$ toad will escape Method 2 •² $\frac{32}{1-\frac{1}{3}}$ or $L = \frac{1}{3}L + 32$ •³ 48 •⁴ $49 \cdot 803 \dots$ •⁵ $50 \cdot 352 \dots$ •⁶ $50 \cdot 352 > 50 \therefore$ toad will escape Method 3 •² numerical strategy •³ $30 \cdot 0625$ •⁴ $49 \cdot 803 \dots$ •⁵ $50 \cdot 352 \dots$ •⁶ $50 \cdot 352 > 50 \therefore$ toad will escape Method 4 •² & •³ $\frac{13}{1-\frac{3}{4}}$ or $L = \frac{3}{4}L + 13$ •⁴ & •⁵ 52 •⁶ $52 > 50 \therefore$ toad will escape	

5

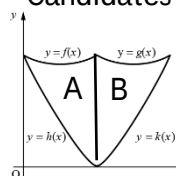
Notes:			
3. • ⁶ is unavailable for candidates who do not consider the toad in their conclusion.			
4. For candidates who only consider the frog numerically award 1/5 for the strategy.			
Commonly Observed Responses:			
Error with frogs limit - Frog Only	Using Method 3 - Toad Only	Using Method 3 - Toad Only	Using Method 3 - Toad Only
$L_F = \frac{34}{1 - \frac{1}{3}}$ $L_F = 51$ $51 > 50$ $\therefore \text{frog will escape.}$	$L_F = 50.352$ $50.352 > 50$ $\therefore \text{the toad escapes.}$	$L_F = 50.1$ $50.1 > 50$ $\therefore \text{the toad escapes.}$	$L_F = 49.7$ $49.7 > 50$ $\therefore \text{the toad escapes.}$
Toad Conclusions Limit = 52 This is greater than the height of the well and so the toad will escape - award • ⁶ . However Limit = 52 and so the toad escapes - • ⁶ ^ .			
Iterations $f_1 = 32$ $f_2 = 42.667$ $f_3 = 46.222$ $f_4 = 47.407$ $f_5 = 47.802$ $f_6 = 47.934$ $f_7 = 47.978$ $f_8 = 47.993$ $f_9 = 47.998$	$t_1 = 13$ $t_2 = 22.75$ $t_3 = 30.0625$ $t_4 = 35.547$ $t_5 = 39.660$ $t_6 = 42.745$ $t_7 = 45.059$ $t_8 = 46.794$ $t_9 = 48.096$ $t_{10} = 49.072$ $t_{11} = 49.804$ $t_{12} = 50.353$		

Question	Generic Scheme	Illustrative Scheme	Max Mark
4 (b)			
•³ know to integrate •⁴ interpret limits •⁵ use 'upper – lower' •⁶ integrate •⁷ substitute limits •⁸ evaluate area between $f(x)$ and $h(x)$ •⁹ state total area	•³ $\int$ •⁴ $\int_0^2$ •⁵ $\int_0^2 (\frac{1}{4}x^2 - \frac{1}{2}x + 3) - (\frac{3}{8}x^2 - \frac{9}{4}x + 3) dx$ •⁶ $-\frac{1}{24}x^3 + \frac{7}{8}x^2$ accept unsimplified integral •⁷ $(-\frac{1}{24} \times 2^3 + \frac{7}{8} \times 2^2) - 0$ •⁸ $\frac{19}{6}$ •⁹ $\frac{19}{3}$	7	
Notes:			
2. If limits $x = 0$ and $x = 2$ appear ex nihilo award •⁴. 4. If a candidate differentiates at •⁶ then •⁶, •⁷ and •⁸ are not available. However, •⁹ is still available. 5. Candidates who substitute at •⁷, without attempting to integrate at •⁶, cannot gain •⁶, •⁷ or •⁸. However, •⁹ is still available. 6. Evidence for •⁸ may be implied by •⁹. 7. •⁹ is a strategy mark and should be awarded for correctly multiplying their solution at •⁸, or for any other valid strategy applied to previous working. 8. For •⁵ both a term containing a variable and the constant term must be dealt with correctly. 9. In cases where •⁵ is not awarded, •⁶ may be gained for integrating an expression of equivalent difficulty ie a polynomial of at least degree two. •⁶ is unavailable for the integration of a linear expression. 10. •⁸ must be as a consequence of substituting into a term where the power of x is not equal to 1 or 0.			

Commonly Observed Responses:

Candidate A - Valid Strategy

Candidates who use the strategy:



Total Area = Area A + Area B

Then mark as follows:

Mark Area A for $\bullet^3$ to $\bullet^8$ then mark Area B for $\bullet^3$ to $\bullet^8$ and award the higher of the two. $\bullet^9$ is available for correctly adding two equal areas.

Candidate B - Invalid Strategy

For example, candidates who integrate each of the four functions separately within an invalid strategy

$\bullet^3$ ✓

Gain $\bullet^4$ if limits correct on

$$\int f(x) \text{ and } \int h(x)$$

or

$$\int g(x) \text{ and } \int k(x)$$

$\bullet^5$ is unavailable

Gain $\bullet^6$ for calculating either

$$\int f(x) \text{ or } \int g(x)$$

and

$$\int h(x) \text{ or } \int k(x)$$

Gain $\bullet^7$ for correctly substituting at least twice

Gain $\bullet^8$ for evaluating at least two integrals correctly

$\bullet^9$ is unavailable

Candidate C

$$\int_0^2 \left(\frac{1}{4}x^2 - \frac{1}{2}x + 3 - \frac{3}{8}x^2 - \frac{9}{4}x + 3 \right) dx$$

$$\int_0^2 \left(-\frac{1}{8}x^2 - \frac{11}{4}x \right) dx \quad \bullet^5 \checkmark$$

$$\frac{-1}{24}x^3 - \frac{11}{8}x^2 \quad \bullet^6 \times$$

Candidate D

$$\int_0^2 \left(\frac{1}{4}x^2 - \frac{1}{2}x + 3 - \frac{3}{8}x^2 - \frac{9}{4}x + 3 \right) dx$$

$$\int_0^2 \left(-\frac{1}{8}x^2 - \frac{11}{4}x + 6 \right) dx \quad \bullet^5 \times$$

$$-\frac{1}{24}x^3 - \frac{11}{8}x^2 + 6x \quad \bullet^6 \boxed{\checkmark 1}$$

Candidate E

$$\int \dots = -\frac{1}{3} \text{ cannot be negative so } = \frac{1}{3} \bullet^8 \times$$

$$\text{however, } = -\frac{1}{3} \text{ so Area } = \frac{1}{3} \bullet^8 \checkmark$$

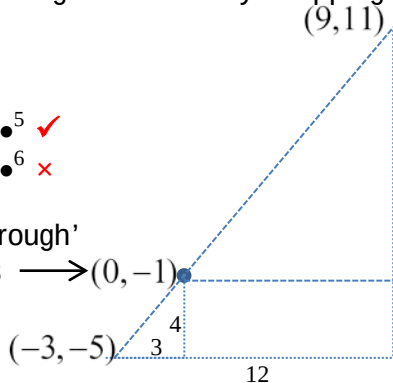
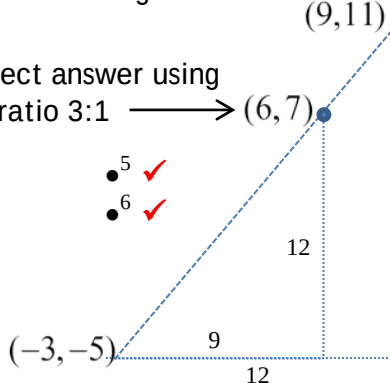
Candidate F

$$\int_0^2 \left(\frac{1}{4}x^2 - \frac{1}{2}x + 3 - \frac{3}{8}x^2 - \frac{9}{4}x + 3 \right) dx$$

$$\int_0^2 \left(-\frac{1}{8}x^2 + \frac{7}{4}x \right) dx \quad \bullet^5 \checkmark$$

$$-\frac{1}{24}x^3 + \frac{7}{8}x^2 \quad \bullet^6 \checkmark$$

Question	Generic Scheme	Illustrative Scheme	Max Mark
5(a)			
• ¹ state centre of C ₁	• ¹ (-3,-5)	4	
• ² state radius of C ₁	• ² 5		
• ³ calculate distance between centres of C ₁ and C ₂	• ³ 20		
• ⁴ calculate radius of C ₂	• ⁴ 15		
Notes:			
1. For • ⁴ to be awarded radius of C ₂ must be greater than the radius of C ₁ .			
2. Beware of candidates who arrive at the correct solution by finding the point of contact by an invalid strategy.			
3. • ⁴ is for Distance _{c1c2} - r _{c1} but only if the answer obtained is greater than r _{c1} .			
Commonly Observed Responses:			

Question	Generic Scheme	Illustrative Scheme	Max Mark
5 (b)			
•⁵ find ratio in which centre of C_3 divides line joining centres of C_1 and C_2 •⁶ determine centre of C_3 •⁷ calculate radius of C_3 •⁸ state equation of C_3	•⁵ 3:1 •⁶ (6,7) •⁷ $r = 20$ (answer must be consistent with distance between centres) •⁸ $(x-6)^2 + (y-7)^2 = 400$	4	
Notes:			
4. For •⁵ accept ratios $\pm 3:\pm 1$, $\pm 1:\pm 3$, $\mp 3:\pm 1$, $\mp 1:\pm 3$ (or the appearance of $\frac{3}{4}$). 5. •⁷ is for $r_{c_2} + r_{c_1}$. 6. Where candidates arrive at an incorrect centre or radius from working then •⁸ is available. However •⁸ is not available if either centre or radius appear ex nihilo (see note 5). 7. Do not accept 20^2 for •⁸. 8. For candidates finding the centre by 'stepping out' the following is the minimum evidence for •⁵ and •⁶:			
•⁵ ✓ •⁶ ✗ Correct 'follow through' using the ratio 1:3 → (0, -1)  Correct answer using the ratio 3:1 → (6, 7) •⁵ ✓ •⁶ ✓ 			
Commonly Observed Responses:			
Candidate A using the mid-point of centres: •⁵ ✗ centre $C_3 = (3, 3)$ •⁶ ✓ 2 radius of $C_3 = 20$ •⁷ ✓ $(x-3)^2 + (y-3)^2 = 400$ •⁸ ✓ 1	Candidate B $C_1 = (-3, -5)$ → $C_2 (9, 11)$ $r = 20$ 1:3 $C_3 = \frac{1}{4} \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ •⁵ ✓ note 4 $C_3 = (0, -1)$ •⁶ ✓ 2 $x^2 + (y+1)^2 = 400$ •⁷ ✓ •⁸ ✓ 1		
Candidate C - touches C_1 internally only •⁵ ✗ •⁶ centre $C_3 = (3, 3)$ ✗ •⁷ radius of $C_3 = \text{radius of } C_2 = 15$ ✓ 1 •⁸ $(x-3)^2 + (y-3)^2 = 225$ ✓ 1	Candidate D - touches C_2 internally only •⁵ ✗ •⁶ centre $C_3 = (3, 3)$ ✗ •⁷ radius of $C_3 = \text{radius of } C_1 = 5$ ✓ 1 •⁸ $(x-3)^2 + (y-3)^2 = 25$ ✓ 1		
Candidate E - centre C_3 collinear with C_1, C_2 •⁵ ✗ •⁶ e.g. centre $C_3 = (21, 27)$ ✗ •⁷ radius of $C_3 = 45$ (touch C_1 internally only) ✓ 1 •⁸ $(x-21)^2 + (y-27)^2 = 2025$ ✓ 1			

Question	Generic Scheme	Illustrative Scheme	Max Mark
6 (a)			
•¹ Expands •² Evaluate p.q •³ Completes evaluation	•¹ p.q + p.r •² $4\frac{1}{2}$ •³ $\dots + 0 = 4\frac{1}{2}$	3	
Notes:			
1. For p.(q + r) = pq + pr with no other working • ¹ is not available.			
Commonly Observed Responses:			
6 (b)			
•⁴ correct expression	•⁴ -q + p + r or equivalent	1	
6 (c)			
•⁵ correct substitution •⁶ start evaluation •⁷ find expression for r 	•⁵ -q.q + q.p + q.r •⁶ $-9 + \dots + 3 \mathbf{r} \cos 30^\circ = 9\sqrt{3} - \frac{9}{2}$ •⁷ $\mathbf{r} = \frac{3\sqrt{3}}{\cos 30}$	3	
Notes:			
2. Award • ⁵ for -q² + q.p + q.r			
Commonly Observed Responses:			
Candidate A		Candidate B	
$-\mathbf{q.q + q.p + q.r} = 9\sqrt{3} - \frac{9}{2}$$-9 + \frac{9}{2} + 3 \mathbf{r} \cos 150^\circ = 9\sqrt{3} - \frac{9}{2}$$\mathbf{r} = \frac{3\sqrt{3}}{\cos 150}$ •⁵ ✓ •⁶ ✗ •⁷ ✓1		$-\mathbf{q.q + q.p + q.r} = 9\sqrt{3} - \frac{9}{2}$$-9 + \frac{9}{2} + 3 \mathbf{r} \cos 30^\circ = 9\sqrt{3} - \frac{9}{2}$$\mathbf{r} = 6$ •⁵ ✓ •⁶ ✓ •⁷ ✓	

Question	Generic Scheme	Illustrative Scheme	Max Mark
7 (a)			
•¹ integrate a term •² complete integration with constant	•¹ $\frac{3}{2} \sin 2x$ OR x •² $x + c$	•¹ $\frac{3}{2} \sin 2x + c$	2
Notes:			
Commonly Observed Responses:			
7 (b)			
•³ substitute for $\cos 2x$ •⁴ substitute for 1 and complete	•³ $3(\cos^2 x - \sin^2 x) \dots$ or $\dots(\sin^2 x + \cos^2 x)$ •⁴ $\dots(\sin^2 x + \cos^2 x) = 4\cos^2 x - 2\sin^2 x$		2
Notes:			
1. Any valid substitution for $\cos 2x$ is acceptable for • ³ . 2. Candidates who show that $4\cos^2 x - 2\sin^2 x = 3\cos 2x + 1$ may gain both marks. 3. Candidates who quote the formula for $\cos 2x$ in terms of A but do not use in the context of the question cannot gain • ³ .			
Commonly Observed Responses:			
Candidate A $3\cos 2x + 1 = (2\cos^2 x - 1) + (2\cos^2 x - 1) + (1 - 2\sin^2 x) + 1$ •³ ✓ •⁴ ✓ $= 4\cos^2 x - 2\sin^2 x$			
Candidate B $4\cos^2 x - 2\sin^2 x = 2(\cos 2x + 1) - (1 - \cos 2x)$ •³ ✓ •⁴ ✓ $= 3\cos 2x + 1$			
7 (c)			
•⁵ interpret link •⁶ state result	•⁵ $-\frac{1}{2} \int \dots$ •⁶ $-\frac{3}{4} \sin 2x - \frac{1}{2} x + c$		2
Notes:			
Commonly Observed Responses:			
Candidate A $\int \sin^2 x - 2\cos^2 x \, dx$ $= \int (3\cos 2x + 1) \, dx$ •⁵ ✗ $\frac{3}{2} \sin 2x + x + c$ •⁶ ✗			

Question	Generic Scheme	Illustrative Scheme	Max Mark
8 (a) (i)			
• ¹ calculate T when $x = 20$	• ¹ 10·4 or 104	1	
8 (a) (ii)			
• ² calculate T when $x = 0$	• ² 11 or 110	1	
Notes:			
1. Accept correct answers with no units. 2. Accept $5\sqrt{436}$ or $10\sqrt{109}$ or equivalent for $T(20)$. 3. For correct substitution alone, with no calculation • ¹ and • ² are not available. 4. For candidates who calculate T when $x = 0$ at • ¹ then • ² is available as follow through for calculating T when $x = 20$ in part(ii).			
Commonly Observed Responses:			
a)	(i) 10·4 • ¹ ✓ See note 1 (ii) 110 • ² ✓		
b)	leading to 9·8seconds • ¹⁰ ✗ See note 7		

Question	Generic Scheme	Illustrative Scheme	Max Mark
8 (b)			
	•³ write function in differential form •⁴ start differentiation of first term •⁵ complete differentiation of first term •⁶ complete differentiation and set candidate's derivative = 0 •⁷ start to solve •⁸ know to square both sides •⁹ find value of x •¹⁰ calculate minimum time	•³ $5(36 + x^2)^{\frac{1}{2}} + \dots$ •⁴ $5 \times \frac{1}{2} ()^{-\frac{1}{2}} \dots$ •⁵ $\times 2x$..... •⁶ $5x(36 + x^2)^{-\frac{1}{2}} - 4 = 0$ $5x = 4(36 + x^2)^{\frac{1}{2}}$ •⁷ or $\frac{5x}{(36 + x^2)^{\frac{1}{2}}} = 4$ $25x^2 = 16(36 + x^2)$ •⁸ or $\frac{25x^2}{(36 + x^2)} = 16$ •⁹ $x = 8$ •¹⁰ $T = 9 \cdot 8$ or 98 no units required	8
Notes:			
5. Incorrect expansion of $(\dots)^{\frac{1}{2}}$ at stage •³ only •⁶ and •¹⁰ are available as follow through. 6. Incorrect expansion of $(\dots)^{-\frac{1}{2}}$ at stage •⁷ only •¹⁰ is available as follow through. 7. Where candidates have omitted units, then •¹⁰ is only available if the implied units are consistent throughout their solution. 8. •¹⁰ is only available as a follow through for a value which is less than the values obtained for the two extremes.			
Commonly Observed Responses:			

Question	Generic Scheme	Illustrative Scheme	Max Mark								
9.											
•¹ use compound angle formula •² compare coefficients •³ process for k •⁴ process for a •⁵ equates expression for h to 100 •⁶ write in standard format and attempt to solve •⁷ solve equation for $1.5t$ •⁸ process solutions for t	•¹ $k \sin 1.5t \cos a - k \cos 1.5t \sin a$ •² $k \cos a = 36, k \sin a = 15$ stated explicitly •³ $k = 39$ •⁴ $a = 0.39479 \dots \text{rad or } 22.6^\circ$ •⁵ $39 \sin(1.5t - 0.39479 \dots) + 65 = 100$ •⁶ $\sin(1.5t - 0.39479 \dots) = \frac{35}{39}$ $\Rightarrow 1.5t - 0.39479 \dots = \sin^{-1}\left(\frac{35}{39}\right)$ <table><tr><td>•⁷</td><td></td><td>•⁸</td></tr><tr><td>1.5t = 1.508</td><td>and</td><td>2.422</td></tr><tr><td>•⁸ t = 1.006</td><td>and</td><td>1.615</td></tr></table>	• ⁷		• ⁸	1.5t = 1.508	and	2.422	• ⁸ t = 1.006	and	1.615	8
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Notes:											
1. Treat $k \sin 1.5t \cos a - \cos 1.5t \sin a$ as bad form only if the equations at the •² stage both contain k. 2. $39 \sin 1.5t \cos a - 39 \cos 1.5t \sin a$ or $39(\sin 1.5t \cos a - \cos 1.5t \sin a)$ is acceptable for •¹ and •³. 3. Accept $k \cos a = 36$ and $-k \sin a = -15$ for •². 4. •² is not available for $k \cos 1.5t = 36$ and $k \sin 1.5t = 15$, however, •⁴ is still available. 5. •³ is only available for a single value of $k, k > 0$. 6. •⁴ is only available for a single value of a. 7. The angle at •⁴ must be consistent with the equations at •² even when this leads to an angle outwith the required range. 8. Candidates who identify and use any form of the wave equation may gain •¹, •² and •³, however, •⁴ is only available if the value of a is interpreted for the form $k \sin(1.5t - a)$. 9. Candidates who work consistently in degrees cannot gain •⁸. 10. Do not penalise additional solutions at •⁸. 11. On this occasion accept any answers which round to 1.0 and 1.6 (2 significant figures required).											

Commonly Observed Responses:

Response 1: Missing information in working.

Candidate A	Candidate B	Candidate C
$39\cos a = 36$ $-39\sin a = -15$ $\tan a = \frac{15}{36}$ $a = 0.39479\dots\text{rad or } 22.6^\circ$	$\cos a = 36$ $\sin a = 15$ $\tan a = \frac{15}{36}$ $a = 0.39479\dots\text{rad or } 22.6^\circ$ Does not satisfy equations at $\bullet^2$	$k \sin 1.5t \cos a - k \cos 1.5t \sin a$ $k \cos a = 36, k \sin a = 15$ $k = 39 \text{ or } -39$ $\tan a = \frac{15}{36}$ $a = 0.39479\dots\text{rad or } 22.6^\circ$ or $a = 3.53638\dots\text{rad or } 202.6^\circ$

Response 2: Correct expansion of $k \sin(x + a)^\circ$ and possible errors for $\bullet^2$ and $\bullet^4$

Candidate D	Candidate E	Candidate F
$k \cos a = 36$ $k \sin a = 15$ $\tan a = \frac{36}{15}$ $a = 1.176\dots\text{rad or } 67.4^\circ$	$k \cos a = 15$ $k \sin a = 36$ $\tan a = \frac{36}{15}$ $a = 1.176\dots\text{rad or } 67.4^\circ$	$k \cos a = 36$ $k \sin a = -15$ $\tan a = \frac{-15}{36}$ $a = 5.888\dots\text{rad or } 337.4^\circ$

Response 3: Labelling incorrect, $\sin(A - B) = \sin A \cos B - \cos A \sin B$ from formula list.

Candidate G	Candidate H	Candidate I
$k \sin A \cos B - k \cos A \sin B$ $k \cos a = 36$ $k \sin a = 15$ $\tan a = \frac{15}{36}$ $a = 0.39479\dots\text{rad or } 22.6^\circ$	$k \sin A \cos B - k \cos A \sin B$ $k \cos 1.5t = 36$ $k \sin 1.5t = 15$ $\tan 1.5t = \frac{15}{36}$ $1.5t = 0.39479\dots\text{rad or } 22.6^\circ$	$k \sin A \cos B - k \cos A \sin B$ $k \cos B = 36$ $k \sin B = 15$ $\tan B = \frac{15}{36}$ $B = 0.39479\dots\text{rad or } 22.6^\circ$

Candidate J	Candidate K
$39 \sin(1.5t - 0.395) = 100$ $\sin(1.5t - 0.395) = \frac{100}{39}$ $1.5t - 0.395 = \sin^{-1} \frac{100}{39}$	$39 \sin(1.5t - 0.395) = 100$ $1.5t - 0.395 = \sin^{-1} \frac{39}{100}$

[END OF MARKING INSTRUCTIONS]